中关村主干网并行网络设计方案

1. 网络资源

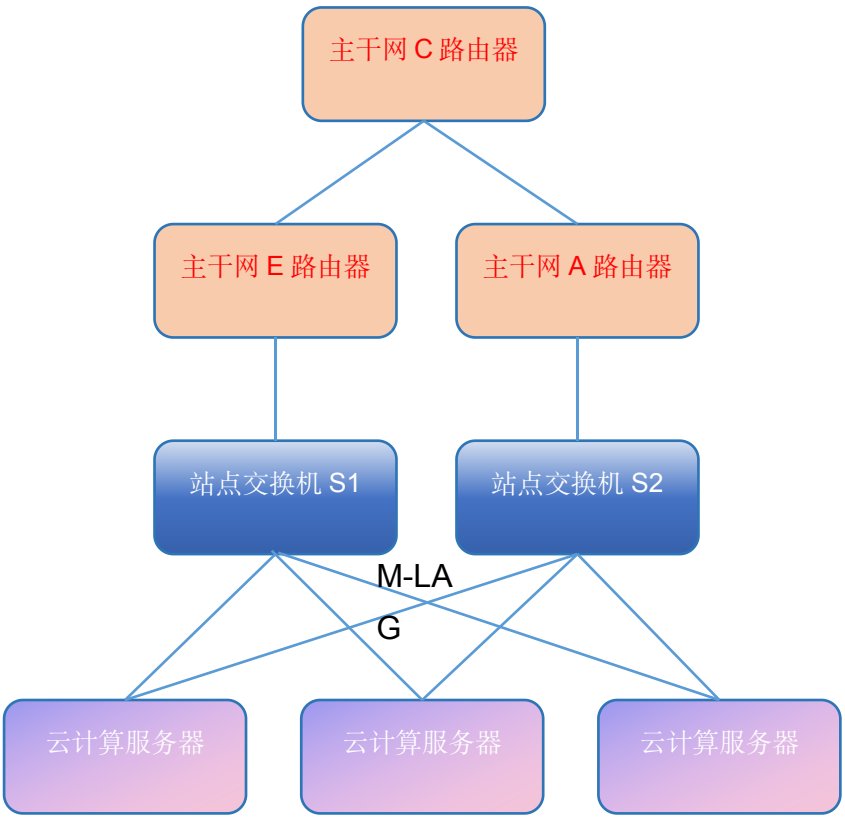
1.1. 公网 IPv6 地址

240a:a806::/32 (从 FITI-G 借用，切片 2055)
240a:a807::/32 （切片 2056）

1.2. VLAN

定制并行网络	3400-3500
L2VPN+L3VPN	3501-3900
虚拟化自用	2000-3000

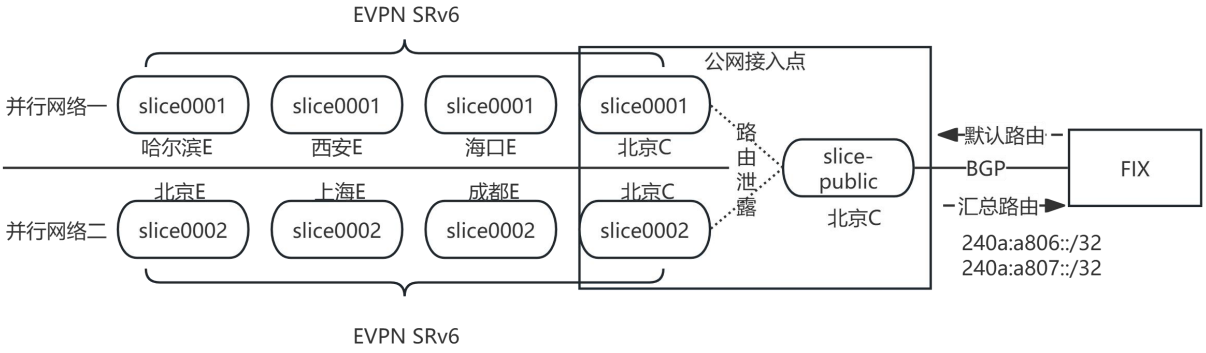
2. 站点网络拓扑



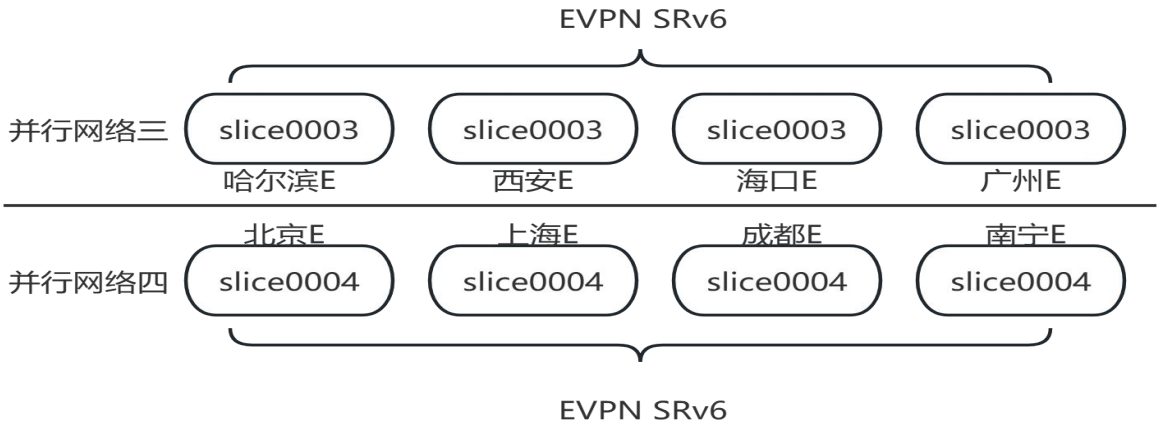
所有并行网络由 E 路由器（试验路由器）承载。S1 上行放行并行网络 VLAN，S2 上行拒绝。

3. 配置图示

3.1. L3VPN



3.2. L2VPN



4. 地址 VLAN 规划

ID	VLAN	站点 ID-0	站点 ID-1	...	站点 ID-40
1	3501	240a:a806:1:0::/64 240a:a806:1:100::/64	240a:a806:1:1::/64 240a:a806:1:101::/64	...	240a:a806:1:40::/64 240a:a806:1:140::/64
2	3502	240a:a806:2:0::/64 240a:a806:2:100::/64	240a:a806:2:1::/64 240a:a806:2:101::/64	...	240a:a806:2:40::/64 240a:a806:2:140::/64
3	3503	240a:a806:3:0::/64 240a:a806:3:100::/64	240a:a806:3:1::/64 240a:a806:3:101::/64	...	240a:a806:3:40::/64 240a:a806:3:140::/64
4	3504	240a:a806:4:0::/64 240a:a806:4:100::/64	240a:a806:4:1::/64 240a:a806:4:101::/64	...	240a:a806:4:40::/64 240a:a806:4:140::/64
....	....	....	....	...	....
399	3899	240a:a806:399:0::/64 240a:a806:399:100::/64	240a:a806:399:1::/64 240a:a806:399:101::/64	...	240a:a806:399:40::/64 240a:a806:399:140::/64
400	3900	240a:a806:400:0::/64 240a:a806:400:100::/64	240a:a806:400:1::/64 240a:a806:400:101::/64	...	240a:a806:400:40::/64 240a:a806:400:140::/64

并行网络 ID 的范围 = 1 ~ 400

VLAN = 并行网络 ID + 3500

IP 分为业务 IP 和互联 IP

业务 IP = 240a:a806:{并行网络 ID}:{站点 ID}::/64

互联 IP = 240a:a806:{并行网络 ID}:{100 + 站点 ID}::/64

主干网侧：240a:a806:{并行网络 ID}:{100 + 站点 ID}::1/64

用户侧：240a:a806:{并行网络 ID}:{100 + 站点 ID}::2/64

5. 并行网络配置举例

5.1. L3VPN（并行网络 ID=300）

5.1.1.试验（E）路由器

所有站点的 E 路由器上均需下发此配置，本处以上海站点（站点 ID=2）为示例

```
ip vpn-instance zslice0300
 address-family ipv6
```

```
  route-distinguisher 9425:300
  vpn-target 9425:300
  apply-label per-instance

interface 100GE3/0/40.3800
 vlan-type dot1q 3800
 ip binding vpn-instance zslice0300
 ipv6 address 240A:A806:300:102::1/64

ipv6 route-static vpn-instance zslice0300 240a:a806:300:2:: 64
240a:a806:300:2::2 tag 300

ipv6 prefix-list zslice0300 index 10 permit 240a:a806:300:2:: 64
route-policy zslice0300 permit node 10
if-match ipv6 address prefix-list zslice0300

bgp 9425
 ip vpn-instance zslice0300
  address-family ipv6 unicast
   import-route direct
   import-route static route-policy zslice0300
   segment-routing ipv6 locator sh-e1 evpn
   segment-routing ipv6 best-effort evpn
```

5.1.2.核心（C）路由器

此配置仅在北京站点核心路由器上下发

```
ip vpn-instance zslice0300
 address-family ipv6
  route-distinguisher 9425:300
  vpn-target 9425:300
  vpn-target 9425:39000 export-extcommunity
  vpn-target 9425:39001 import-extcommunity
  apply-label per-instance
```

```
bgp 9425
 ip vpn-instance zslice0300
  address-family ipv6 unicast
   segment-routing ipv6 locator bj-c1 evpn
   segment-routing ipv6 best-effort evpn
```

5.2. L2VPN（并行网络 ID=300）

5.2.1.试验（E）路由器

所有站点的 E 路由器上均需下发此配置，本处以上海站点（站点 ID=2）为示例

```
vsi zslice0300
  evpn encapsulation srv6
      route-distinguisher 9425:300
      vpn-target 9425:300
      segment-routing ipv6 best-effort
      segment-routing ipv6 locator sh-e1

interface ten-gigabitethernet 3/1/1.3800
  vlan-type dot1q vid 3800
  xconnect vsi zslice0300
```

6. 基础网络的配置

试验路由器的基础网络已拉通，本处不在赘述。

北京核心路由器 C2 作为并行网络的公网接入点，需要进行以下配置

```
ip vpn-instance zslicefix
 address-family ipv6
  vpn-target 9425:39000 import-extcommunity
  vpn-target 9425:39001 export-extcommunity
```

```
interface HundredGigE1/1/4/1.2055
 ip binding vpn-instance zslicefix
 mac-address 9023-b4a7-a806
 vlan-type dot1q vid 20 second-dot1q 7
 ipv6 address 2001:255:0:100::A806/64

interface HundredGigE1/3/2/1.2055
 vlan-type dot1q vid 20 second-dot1q 7
 xconnect vsi 100

ipv6 prefix-list zslicefromfix index 10 permit :: 0
ipv6 prefix-list zslicetofix index 10 permit 240a:a806:: 32
ipv6 prefix-list zslicetofix index 20 permit 240a:a807:: 32

bgp 9425
ip vpn-instance zslicefix
  router-id 1.18.7.1   (对应切片 2055)
  peer 2001:255:0:100::2000 as-number 38255
  peer 2001:255:0:100::2000 fake-as 144704
  address-family ipv6 unicast
    peer 2001:255:0:100::2000 enable
    peer 2001:255:0:100::2000 prefix-list zslicefromfix import
    aggregate 240a:a806:: 32 detail-suppressed
    aggregate 240a:a807:: 32 detail-suppressed
```